

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 1 of 9

(1) Research Program/Project/Study Title: Development and Quality Evaluation of Sweetpotato Crouton as Affected by Varying Levels of Sweetpotato Flour Substitute	
Proponent Name:	Clyde Dwynehl G. Leccio, Adelfo Z. Virtudazo, Marife R. Hilapad, Lecel O. Bobier, Rhodora L. Ticar, Edzil B. Lorenzo, Virigina P. Cabales, Merlyn F. Filosofo, Lillibeth G. Leonor
Academic Rank:	Instructor I
Co-Authors (with academic rank):	
Campus/Satellite College/Address:	CAPSU, Poblacion Proper, Mambusao, Capiz
Telephone/Cell Phone/Fax Number:	09383293538
E-mail Address:	cdogarcia@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Mambusao
		Province:	Capiz
		Region:	Region 6
(5) Classification		(6) Mode of Implementation	
		☐ Single Authorship/Proponent	

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 2 of 9

Research: ☐ Basic ☒ Applied	Development: ☒ Pilot Testing ☐ Tech. Promotion/Commercialization	☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Food and Agriculture / Sweetpotato		(8) Discipline/Field of Specialization Food Science and Technology
(9) Rationale Sweetpotato is a known to be a staple food in the Philippines. It is one of the rootcrops in the Philippines wherein research efforts were put into so as to develop innovative products out of it as it is known to be very beneficial to the body due to its high antioxidant levels which is good for the body and immune system. Sweetpotato now is being continuously studied on since researchers want to add value to the product so as to extend its shelf-life so that it can be stored on the shelves and therefore be available to consumers for a longer period of time. Sweetpotato has a potential to be another variety of crouton. A crouton is a piece of bread that have been toasted, either in a frying pan or the oven, until they are crisp and golden brown. One popular variety of it is the potato crouton.		
(9-A) Project Description The project has a one year duration for developing a sweetpotato product that is more shelf-stable called crouton. Crouton is basically a toasted bread added with flavors like fats and herbs. The project will be composed of product development stage wherein the acceptable formulation is aimed and analysis stage including physico-chemical, microbial and sensory evaluation will be conducted.		
(10) Objectives/Statement of the Problem Generally, this project aims to develop a more shelf-stable product out of sweetpotato and therefore add value to the rootcrop Specifically, the project aims to 1. Identify the appropriate formulation of ingredients for sweetpotato crouton with varying levels of sweetpotato 2. Assess the physico-chemical attributes of sweetpotato crouton in terms of its pH, TSS, and TA as well as the microbial attributes of the product 3. Assess the sensory attribute of the product and identify the most preferred product by the consumers through consumer testing 4. Identify the cost of production in developing sweetpotato crouton.		
(11) Review of Literature/Prior Art Search The sweetpotato, <i>Ipomoea batatas</i> L. (Lam.), is an important economic crop in many countries. It yields reasonably well, even in soils of low fertility. In terms of annual production, sweetpotato ranks as the fifth most important food crop in the tropics and the seventh in the world food production after wheat, rice, maize, potato, barley, and cassava (FAO 2016). Sweetpotato fulfills a		

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 3 of 9

number of basic roles in the global food system, all of which have fundamental implications for meeting food requirements, reducing poverty, and increasing food security (El-Sheikha and Ray 2017).

Sweetpotato production in the Philippines. Eastern Visayas posted the largest production with 46.81 thousand metric tons, contributing 34.5 percent to the country's total output this quarter. The other top producing regions were Bico region and Caraga with 10.9 and 9.7 percent share, respectively. However, the Philippine Statistics Authority's 2021 report that during the third quarter of 2021, production of sweetpotato was registered at 135.74 thousand metric tons. It dropped by -1.4%, from 137.70 thousand metric tons output in the same quarter of 2020.

Nutritional Content and Value of Sweetpotato

The sweet potato (*Ipomoea batatas*) is an underground tuber. It is a nutritious food for humans and animals (Woolfe, 1992; Yamakawa & Yoshimoto, 2001). It is a source of carbohydrate, rich in vitamin A and C (especially in yellow and orange flesh cultivars, but orange cultivars are much higher than yellow flesh cultivars), minerals, especially iron (Fe), phosphorus (P), and calcium (Ca). Sweet potato also contains proteins and fats in low concentrations in the tubers, while the leaves are rich in protein, vitamins and minerals (Woolfe, 1992).

Depending on the flesh color, sweetpotatoes contain high levels of β -carotene, anthocyanins, and phenolics. The β -carotene in orange-fleshed sweetpotatoes can play a significant role as a viable long-term food-based strategy for combating vitamin A deficiency in the world. Studies in Africa demonstrated that increased consumption of orange-fleshed sweetpotatoes improved the vitamin A status of children, pregnant women, and lactating mothers (Low et al. 2007; Hotz et al. 2012; Van Jaarsveld et al. 2005). Further, polyphenolics from purple-fleshed sweetpotatoes exhibited diverse biological activities, including protective antioxidant, anti-inflammatory, anticancer, antidiabetic and hepatoprotective activity effects (Hu et al. 2016; Lim et al. 2013). Sweetpotato has a strong potential to contribute to better nutritional quality of our diets around the world.

It's rich in an antioxidant called beta carotene, which is very effective at raising blood levels of vitamin A, particularly in children. Sweet potatoes are nutritious as they are high in fiber.

Priot Art Search

The patent JP2016000019A authored by Hirosawa Hidejiro and Arai Masahiro entitled CROUTON AND METHOD FOR PRODUCING THE SAME. In this patent, the method for producing a crouton includes a dough that is fired and is shredded and dried to produce a crouton. The raw material for cereal flour for producing a dough using cereals flours blended with (1) starch with a sediment area of 2 or lower in 55 to 60 wt % soybean flour in 10 to 15 wt%, (2) soybean flour in 10 to 15 wt.%, (3)wheat protein in 15 to 20 wt% and (4) wheat flour in 10 to 15 wt%.

Another patent JP3411409B2 entitled A METHOD OF PRODUCING CROUTON authored by Tamaoki Yoji et al. uses potato as a raw material, floating on soap for required time and retaining texture characteristic of crouton. Its consitution is finely cut potato or formed food of powdery potato is subjected to frying treatment at 80-230 oC (under reduced pressure) to produce potato crouton having $\leq 5\text{wt\%}$ of water content and 20-60wt% of oil content.

Matsumura Masahiro et al also authored a METHOD OF PRODUCING CROUTON with patent JP2008067647A in which they wanted to suppress the reduction in the crunching senation of the crouton. The solution includes a first step for attaching melted solid fat to the outside of the bread, the second step for making a powder of the first starch adhere to the bread, and a third step for making the pwoder of the second starch adhere to the bread, when croutons are to be produced.

However, a crouton which is from sweetpotato material hasn't been developed. The sweetpotato has a good potential to be made into a crouton due to its starch content. This will also give a new form of the sweetpotato product as snack foods and or dressing ingredient for salads.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
			Revision No.	00
	Document Title:	RESEARCH PROPOSAL	Effective Date	November 30, 2019
			Page	Page 4 of 9

(12) Scientific Basis/Theoretical Framework

Bread and cakes are considered as intermediate-moisture foods with moisture contents typically ranging from 18% to 25% for cakes and 35–42% for bread. The water activity (a_w) of these products is above 0.95. With such moisture contents and high a_w , baked goods are susceptible to microbial spoilage with the main effect coming from the growth of various molds which is initially manifested by staling.

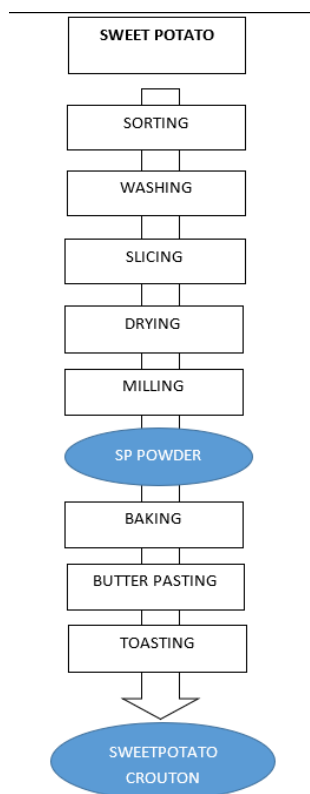
The molecular basis of staling is examined by reviewing what is known about the components of wheat flour, factors that affect staling rate, and the various mechanisms that have been proposed. The conclusion reached is that bread staling is a complex phenomenon in which multiple mechanisms operate. Polymer crystallizations with the formation of supermolecular structures are certainly involved. The most plausible hypothesis is that retrogradation of amylopectin occurs, and because water molecules are incorporated into the crystallites, the distribution of water is shifted from gluten to starch/amylopectin, thereby changing the nature of the gluten network.

(13) Methodology

Procurement of Raw Materials

The main raw material, the sweetpotato, will be procured from the supermarket Mambusao. The sweetpotato will undergo pre-treatment processing prior to processing.

Processing of Sweetpotato Crouton



	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 5 of 9

Figure 1. Sweetpotato Crouton Process Flow

Experimental Design

Variables which will be the bases of development are the levels of sweet potato flour and the amount of butter that will be pasted on the outer crust of the crouton prior to toasting.

A 3x3 factorial design with different levels of sweetpotato and herbs time: 40%, 45%, 50% and 1tsp, 2tsp, 3tsp respectively, is tabulated below. The experimentation will have 9 treatments.

Variables		Levels of Sweetpotato flour (%)		
		40%	45%	50%
Levels of Butter	1 tsp	T ₁	T ₂	T ₃
	2 tsp	T ₄	T ₅	T ₆
	3 tsp	T ₇	T ₈	T ₉

Microbial Analyses

Standard Plate Count/Total Plate Count will be used to enumerate the number of microorganisms.

Physico-chemical Analyses.

Analyses of functional properties of sweetpotato crouton include moisture content analysis. Proximate analyses will also be made.

Sensory Evaluation

Sensory evaluation will be carried out to determine the quality description and acceptability of sweetpotato crouton as affected by sweetpotato levels and herbs. Samples will be presented to 100 panelists composed of Bachelor of Science in Food Technology (BSFT) students at CAPSU, Poblacion Proper, Mambusao, Capiz.

Descriptive Quality Scoring and 9-point Hedonic Scale Scoring will be used.

Shelf-life Analysis

Accelerated shelf-life testing will be used in this experiment to know the shelf stability of the product.

Statistical Analysis

Microbiological data will be presented as growth curves of CFU/g vs time and temperature. Mean values and standard deviation of microbial counts obtained as well as the data in sensory

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 6 of 9

evaluation and functional property analysis will be used for statistical analysis using one-way ANOVA.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Gantt Chart on the Development of Sweetpotato Crouton

Research Objectives	No of Months (Ordinal)											
	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th	9 th	10 th	11 th	12 th
Development of Sweetpotato Crouton												
1. Conduct of Preliminary Studies and Procurement of Raw Materials												
2. Experiment Proper (Production/Baking Process)												
3. Microbial, Functional Property Analysis and Sensory Evaluation												
4. Consumer Testing and Cost Analysis												
5. Statistical Analysis and Paper/Manuscript Writing												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

A new healthy toasted snack food which can be stored in shelves for an extended period of time will be produced upon the completion of the study along with the quality evaluation reports of the product. This product will be beneficial not just to MSME's but also to the consumers.

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 7 of 9

The prime target of developing sweetpotato crouton is its commercialization. The technology should be adopted by an MSME for future commercialization. For this to happen, the following activities are proposed to be done:

- A. Conduct of trainings on Sweetpotato Crouton Processing
- B. Seminar on Proper GMP in Food Processing
- C. Technology Transfer Proper
- D. Commercialization of Sweetpotato Crouton Processing
- E. Quarterly Evaluation/Survey on the Product Reproducibility

(16) Target Beneficiaries

The initial target beneficiaries of the project will be the micro, small, and medium enterprises (MSME's) of Mambusao, Capiz.

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Clyde Dwynehl O. Garcia	Project and Study Leader	Food Science and Technology
Lecel O. Bobier	Proponent	Food Science and Technology

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount/item (PhP)	TOTAL AMOUNT
A. Personnel Services Research Assistant	3months x P11,068.00	P 11,068.00	33,204.00
B. Maintenance and Other Operating Expenses (MOOE)			39,150.00
Experimentation			
Conduct of Preliminary Study			2,500.00
Experiment Proper with Sensory Analysis			10,000.00
Processing materials (chopping board, knife, bowls, strainers, ladles, etc)	2 pcs each		5,000.00
Jars (1L-capacity)	6	200.00	1,200.00
LPG Tank	1	800.00	800.00
Digital Balance	1	500.00	500.00
Microbial Analyses (TPC for LAB)	18 samples		
PCA	500 g	3,000.00	3,000.00
Petri dish	30 pcs	70.00	2,100.00
Test tubes	50	25.00	1,250.00

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
	Document Title:	RESEARCH PROPOSAL	Revision No.	00
			Effective Date	November 30, 2019
			Page	Page 8 of 9

Micro pipettor	1 unit	4,500	4,500.00
Physico-chemical and Proximate Analyses	18 samples	350.00	6,300.00
Travel Expenses		2,000.00	2,000.00
C. CAPITAL OUTLAY			
Furnitures and Fixtures		10,000.00	10,000.00
Total			82,354.00

(19) Literature Cited

AO. 2016. Crop Production Data. Available at <http://www.fao.org/faostat/en/#data/QC> (accessed December 29, 2016).

EL-SHEIKHA AF, RAY RC. 2017. Potential impacts of bioprocessing of sweetpotato: Review. Crit Rev Food Sci & Nutr 57:455–471.

LOW JW, ARIMOND M, OSMAN N, CUNGUARA B, ZANO F, TSCHIRLEY D. 2007. A food-based approach introducing orange-fleshed sweetpotatoes increased vitamin A intake and serum retinol concentrations in young children in rural Mozambique. J Nutr 137: 1320–1327.

WOOLFE, J. A. (1992). Sweet Potato: An Untapped Food Resource. Cambridge University Press. Cambridge, United Kingdom. <http://dx.doi.org/10.12691/ajfst-2-4-1>

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

CLYDE DWYNELH O. GARCIA

Food Technology Department, Capiz State University-Mambusao Satellite College
Email: cdogarcia@capsu.edu.ph

I. EDUCATION

2018 – Master of Science in Food Science and Technology

2012 – Bachelor of Science in Food Technology

II. PROFESSIONAL EXPERIENCE

2021 – Present – Instructor I (Food Technology Department, Capiz State University – Mambusao Satellite College)

2019-2021 – Part-time Instructor (Institute of Food Science and Technology, Cavite State University-Main)

III. AWARDS

2008-2012 – Scholarship Grantee/Recipient | Department of Science and Technology (DOST)

2016-2018 – Scholarship Grantee/Recipient | Commission on Higher Education (CHED)

IV. RESEARCH INTEREST

Research and Development of Functional Foods, Zero-waste processing and utilization of low-economic-value commodities to convert to value-added products, probiotics foods, utilization of high-antioxidant-containing foods

LECEL O. BOBIER

Food Technology Department, Capiz State University-Mambusao Satellite College
Email: lobobier@capsu.edu.ph

I. EDUCATION

	Document Type: DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- RP-01
		Revision No.	00
	Document Title: RESEARCH PROPOSAL	Effective Date	November 30, 2019
		Page	Page 9 of 9

2019 – Doctor of Philosophy in Science Education
2011 – Master of Arts in General Science
1992 – Bachelor of Science in Biological Science

II. PROFESSIONAL EXPERIENCE

2021-Present Associate Professor IV

III. AWARDS

2017-2019 – Grantee/Recipient | Commission on Higher Education (CHED)

IV. RESEARCH INTEREST

Social science researchers